MALLA REDDY UNIVERSITY

An Application Development Project Report

Secure optical vehicle-to-vehicle communication using Li-fi for defence and military vehicle

Submitted by

M. Mani Ranga Reddy	2411CS050027
T. Rakesh	2411CS050062
B. Srikar	2411CS050077
Ch. Bhageerath	2411CS050079
M. Sai Darshan	2411CS050081
S. Mani Chandra	2411CS050083
A. Deepak	2411CS050084
U. Avinash	2411CS050087

GUIDE BY

Dr. Ome Nerella (Associate Professor)

Department of Computer Science & Engineering (Internet of Things)

B.Tech III YEAR - I SEM

Malla Reddy University, Hyderabad

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Department of Computer Science & Engineering (Internet of Things)

CERTIFICATE

This is to certify that this is the bonafide record of the application development entitled **"SECURE OPTICAL VEHICLE-TO-VEHICLE COMMUNICATION USING LI-FI FOR DEFENCE AND MILITARY VEHICLE"** submitted by **M.Mani Ranga Reddy (2411CS050027), T.Rakesh (2411CS050062), B.Srikar (2411CS050077), Ch.Bhageerath (2411CS050079), M.Sai Darshan (2411CS050081), S. Mani Chandra (2411CS050083), A.Deepak (2411CS050084), U.Avinash (2411CS050087)** B. Tech III year I semester, Department of CSE (IoT) during the year 2026-27. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

Internal Guide
Dr. Ome Nerella
(Associate Professor)

Dean (CS, IOT&ECE)
Dr G. Anand Kumar
Professor & Dean

External Examiner

DECLARATION

We hereby declare that the project report entitled "**Secure optical vehicle-to-vehicle communication using li-fi for defence and military vehicle**" has been carried out by us. This work has been submitted to the Department of Computer Science and Engineering (Internet of Things), Malla Reddy University, Hyderabad in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology. we further declare that this project work has not been submitted in full or part for the award of any other degree in other educational institutions.

Place:

Date:

M. Mani Ranga Reddy	2411CS050027
T. Rakesh	2411CS050062
B. Srikar	2411CS050077
Ch. Bhageerath	2411CS050079
M. Sai Darshan	2411CS050081
S. Mani Chandra	2411CS050083
A. Deepak	2411CS050084
U. Avinash	2411CS050087

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M. Mani Ranga Reddy	2411CS050027
T. Rakesh	2411CS050062
B. Srikar	2411CS050077
Ch. Bhageerath	2411CS050079
M. Sai Darshan	2411CS050081
S. Mani Chandra	2411CS050083
A. Deepak	2411CS050084
U. Avinash	2411CS050087

ABSTRACT

Secure Optical Vehicle-to-Vehicle (V2V) Communication Using Light Fidelity (Li-Fi) for Defence and Military Vehicles is a proposed system designed to enable secure, short-range wireless communication between military vehicles using light signals instead of conventional radio-frequency communication. In defence operations, reliable and confidential information exchange between vehicles is essential for improving situational awareness, coordination, and operational safety. Traditional radio-based communication may face challenges such as electromagnetic interference, signal interception, and radio-frequency detection. Li-Fi offers an alternative by transmitting data through modulated light signals, potentially reducing radio-frequency emissions and limiting communication to vehicles within the optical link. The proposed system uses a high-brightness LED as the optical transmitter and a photodiode with an amplifier circuit as the receiver. An ESP32 microcontroller processes sensor information, including acceleration and fire detection, and transmits emergency alerts and predefined messages through Li-Fi. Encryption techniques such as the Advanced Encryption Standard (AES) can be incorporated to protect transmitted information from unauthorized access. At the receiving vehicle, the photodiode converts the received light signals into electrical signals, which are processed by the microcontroller and displayed on an LCD. The system can be demonstrated using two vehicle models to simulate emergency communication and warning-message exchange. This project aims to demonstrate a low-cost, short-range optical communication approach for defence and military vehicle applications. By combining optical data transmission, sensor-based emergency detection, and encryption, the proposed system can support secure vehicle coordination in suitable line-of-sight conditions. However, its practical effectiveness depends on factors such as visibility, alignment, transmission distance, and environmental conditions. Li-Fi is therefore considered a complementary communication technology rather than a complete replacement for existing military communication systems.

Keywords: Li-Fi, Secure Optical Communication, Vehicle-to-Vehicle (V2V) Communication, Defence and Military Vehicles, ESP32 Microcontroller, AES Encryption, Optical Wireless Communication, LED Transmitter, Photodiode Receiver.

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CHAPTER 1

INTRODUCTION

In modern defence and military operations, secure and reliable communication between vehicles is essential for mission coordination, situational awareness, and personnel safety. Military vehicles often operate in challenging environments where communication reliability, information confidentiality, and rapid emergency response are critical. Conventional radio-frequency (RF) communication systems may experience electromagnetic interference and can potentially be intercepted or detected. Therefore, exploring alternative communication technologies that provide secure and efficient data transmission is important for defence applications.

Light Fidelity (Li-Fi) is an optical wireless communication technology that transmits data using light emitted by sources such as light-emitting diodes (LEDs). Unlike conventional RF communication, Li-Fi uses modulated light signals to transmit digital information through an optical channel. A photodiode at the receiving end detects these light signals and converts them into electrical signals, which can then be processed by a microcontroller. Li-Fi offers potential advantages, including reduced RF emissions, localized communication, and the possibility of limiting signal reception to vehicles within the optical link. However, its performance depends on line-of-sight conditions, distance, alignment, and environmental lighting.

The proposed **Secure Optical Vehicle-to-Vehicle (V2V) Communication Using Li-Fi for Defence and Military Vehicles** system aims to demonstrate secure information exchange between two vehicle models using optical signals. The system uses an ESP32 microcontroller, a high-brightness LED transmitter, a photodiode receiver with an amplifier circuit, and suitable sensors to detect conditions such as sudden acceleration changes and fire-related emergencies. Important information, including emergency alerts and predefined messages, is transmitted through the Li-Fi channel. Encryption techniques, such as the Advanced Encryption Standard (AES), can be incorporated to protect the transmitted data from unauthorized interpretation. At the receiving end, the ESP32 processes the received information and displays the message on an LCD.

1.1 Problem Definition & Description

Communication between defence and military vehicles is essential for exchanging critical information, coordinating movements, and responding to emergency situations. Conventional radio-frequency (RF) communication systems may face challenges such as electromagnetic interference, signal interception, and unwanted RF emissions. These challenges create a need to explore alternative communication technologies that support localized and secure information exchange. The proposed project, Secure Optical Vehicle-to-Vehicle (V2V) Communication Using Li-Fi for Defence and Military Vehicles, aims to develop a system that transmits emergency alerts and important messages between vehicles using modulated light signals.

The system uses ESP32 microcontrollers, a high-brightness LED transmitter, a photodiode receiver with an LM358 amplifier circuit, and sensors such as the MPU6050 accelerometer and a fire detection sensor to identify predefined emergency conditions. When an emergency is detected, the transmitter generates an alert message and sends it through the optical communication channel. Encryption techniques such as the Advanced Encryption Standard (AES) can be integrated to protect the confidentiality of transmitted messages, while the receiving ESP32 processes the received data and displays the information on an LCD.

The primary objective is to demonstrate a low-cost optical communication prototype that supports encrypted message transmission and sensor-based emergency alerts while reducing dependence on radio-frequency communication for the optical link. However, the system's performance depends on line-of-sight availability, transmission distance, alignment, and ambient lighting conditions. Therefore, the proposed prototype is intended to demonstrate the feasibility of secure optical communication as a complementary technology for selected defence and military vehicle applications.

1.2 Objectives of the Project

The main objective of this project is to develop a Secure Optical Vehicle-to-Vehicle (V2V) Communication System using Light Fidelity (Li-Fi) for defence and military vehicles. The system aims to enable secure and efficient data transmission between vehicles using modulated light signals instead of conventional radio-frequency communication. It focuses on integrating optical wireless communication, data encryption, and sensor-based emergency detection to improve communication confidentiality and support timely information exchange.

The specific objectives are to establish communication between two vehicles using a high-brightness LED and a photodiode receiver, implement encryption techniques such as AES to protect transmitted messages from unauthorized access, detect emergency conditions using sensors such as the MPU6050 accelerometer and a fire detection sensor, transmit emergency alerts and predefined messages through the Li-Fi channel, and display received information on an LCD using an ESP32 microcontroller.

The project also aims to demonstrate a low-cost prototype, evaluate optical communication under different lighting and distance conditions, and explore Li-Fi as a complementary communication technology for defence and military vehicle applications.

1.3 Scope of the Project

The scope of this project is to design and develop a prototype of Secure Optical Vehicle-to-Vehicle (V2V) Communication Using Li-Fi for Defence and Military Vehicles. The system focuses on enabling short-range communication between two vehicles using a high-brightness LED as the transmitter and a photodiode with an amplifier circuit as the receiver. ESP32 microcontrollers are used to process sensor data, transmit messages, and receive information. The system can detect predefined emergency conditions using sensors such as the MPU6050 accelerometer and a fire detection sensor, allowing emergency alerts and important messages to be transmitted between vehicles.

AES encryption can be integrated to improve data confidentiality and prevent unauthorized interpretation of transmitted messages. An LCD is used to display the received information, making the system easy to monitor during testing. The project can be further enhanced by improving communication range, transmission speed, error detection, authentication, and performance under different environmental conditions.

Future developments may include multi-vehicle communication, advanced collision detection, improved optical receivers, and integration with existing communication technologies. The proposed system is intended as a research prototype to demonstrate the potential of Li-Fi as a complementary communication technology for defence and military applications, rather than a replacement for established military communication systems.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Existing System

In existing Vehicle-to-Vehicle (V2V) communication systems, information is commonly exchanged using radio-frequency (RF) technologies such as Wi-Fi, Bluetooth, and dedicated short-range communication systems. These technologies enable vehicles to share information about their position, speed, traffic conditions, and emergency situations. However, RF-based communication may be affected by electromagnetic interference, signal congestion, and potential interception by unauthorized users. In defence and military applications, unwanted RF emissions may also increase the possibility of communication signals being detected.

Traditional systems may require additional security mechanisms, such as encryption and authentication, to protect sensitive information. Furthermore, some conventional communication systems depend on network infrastructure or specialized equipment, which can increase implementation complexity and cost. These limitations have encouraged research into alternative communication technologies, including Light Fidelity (Li-Fi), which uses modulated light signals to transmit data.

Li-Fi offers the potential for localized communication with reduced dependence on RF transmission, but it requires suitable optical conditions and does not inherently guarantee secure communication. Therefore, an alternative system combining optical transmission, data encryption, and sensor-based emergency alerts can be explored to address selected limitations of existing communication approaches.

2.1.1 Background & Literature review

Vehicle-to-Vehicle (V2V) communication is an important technology in modern transportation and defence applications, enabling vehicles to exchange information about their movements, status, and emergency situations. Conventional communication technologies, such as Dedicated Short-Range Communications (DSRC), Wi-Fi, and Cellular Vehicle-to-Everything (C-V2X), use radio-frequency signals to transmit data but may face challenges related to electromagnetic interference, signal interception, and communication security. Light Fidelity (Li-Fi) has emerged as an alternative optical wireless communication technology that transmits data through modulated LED light signals, which are detected by photodiodes at the receiving end.

Previous research has explored Li-Fi for high-speed, localized communication and applications requiring reduced dependence on radio-frequency transmission. Research on secure communication has also highlighted the importance of encryption algorithms, such as the Advanced Encryption Standard (AES), to protect sensitive information from unauthorized access.

Literature Survey

· **Li-Fi-Based Communication:** Previous studies have explored Light Fidelity (Li-Fi) technology for wireless data transmission using modulated LED light signals and photodiode receivers. Li-Fi offers localized communication with reduced dependence on radio-frequency transmission.
· **Vehicle-to-Vehicle (V2V) Communication:** Research on V2V communication has focused on technologies such as Dedicated Short-Range Communications (DSRC) and Cellular Vehicle-to-Everything (C-V2X) to exchange vehicle information, safety messages, and emergency alerts.
· **Secure Data Transmission:** Studies on communication security have investigated encryption algorithms such as the Advanced Encryption Standard (AES) to protect transmitted information from unauthorized access and interpretation.
· **Sensor-Based Emergency Detection:** Previous research has explored the use of accelerometers and fire detection sensors with microcontrollers to identify abnormal vehicle movements, potential collisions, and fire-related emergencies.

2.1.2 Limitations of Existing System

Existing Vehicle-to-Vehicle (V2V) communication systems primarily use radio-frequency technologies such as Wi-Fi, Dedicated Short-Range Communications (DSRC), and Cellular Vehicle-to-Everything (C-V2X). These systems may face challenges such as electromagnetic interference, signal congestion, potential signal interception, and cybersecurity threats if appropriate protection mechanisms are not implemented. In defence and military applications, radio-frequency emissions may also create concerns regarding the detection of communication signals by adversaries.

Some existing systems require specialized equipment, network infrastructure, or additional security mechanisms, which can increase implementation complexity and cost. Furthermore, conventional communication systems may not always integrate sensor-based emergency detection and secure message transmission in a single low-cost prototype. Although Li-Fi offers an alternative to radio-frequency communication, it also has limitations, including the requirement for a suitable optical path, sensitivity to ambient lighting, limited communication range, and alignment difficulties. These limitations highlight the need to explore an integrated communication system that combines optical data transmission, encryption, and sensor-based emergency alerts for defence and military vehicle applications.

2.2 Proposed System

The proposed system, **Secure Optical Vehicle-to-Vehicle (V2V) Communication Using Li-Fi for Defence and Military Vehicles**, aims to provide a secure and efficient method of exchanging information between vehicles using optical wireless communication. The system uses ESP32 microcontrollers, a high-brightness LED as the transmitter, and a photodiode with an LM358 amplifier circuit as the receiver. Sensors such as the MPU6050 accelerometer and a fire detection

sensor are integrated to identify predefined emergency conditions and generate warning messages. The ESP32 processes the sensor information and transmits emergency alerts and predefined messages through modulated LED light signals. AES encryption can be implemented to protect the transmitted data, while the receiving ESP32 processes the received optical signals, decrypts the message, and displays the information on an LCD. This approach aims to reduce dependence on radio-frequency transmission for the communication link while providing localized data transfer and sensor-based emergency notifications. The proposed system can be demonstrated using two vehicle models to evaluate message transmission, emergency detection, and communication performance under different lighting and distance conditions. However, the system requires a suitable optical path and does not guarantee complete security or replace established military communication systems.

2.2.1 Advantages of the Proposed System

· **Enhanced Data Security:** AES encryption helps protect transmitted messages from unauthorized interpretation.
· **Reduced RF Dependence:** Uses optical signals for communication, reducing the need for radio-frequency transmission on the Li-Fi link.
· **Emergency Alert Generation:** Sensors detect predefined emergency conditions and enable the transmission of warning messages.
· **Localized Communication:** Light-based transmission can help confine communication to the area covered by the optical link.
· **Real-Time Information Exchange:** Enables rapid transmission of emergency alerts and predefined messages between vehicles.
· **Low-Cost Implementation:** Uses affordable components such as ESP32 microcontrollers, LEDs, photodiodes, and sensors.
· **Easy Monitoring:** An LCD displays received messages and emergency alerts for convenient monitoring.
· **Reduced RF Interference:** Optical communication is not directly affected by radio-frequency interference, although ambient light can affect its performance.

2.3 Software & Hardware Requirements

2.3.1 Software Requirements

· **Programming Languages:** Embedded C/C++
· **Microcontroller Programming:** Arduino IDE
· **Microcontroller Platform:** ESP32
· **Communication Technology:** Li-Fi (Optical Wireless Communication)
· **Encryption Algorithm:** AES (Advanced Encryption Standard)

- **Sensor Libraries:** Adafruit MPU6050, Adafruit Unified Sensor
- **LCD Library:** LiquidCrystal_I2C

2.3.2 Hardware Requirements:

- **Microcontroller:** ESP32 Development Board (30-pin, CP2102)
- **Optical Transmitter:** High-Brightness LED
- **Optical Receiver:** BPW34 Photodiode
- **Amplifier Circuit:** LM358 Operational Amplifier Module
- **Accelerometer Sensor:** MPU6050 (Gyroscope and Accelerometer)
- **Fire Detection Sensor:** Flame Sensor Module
- **Display Unit:** 16×2 LCD with I2C Module
- **Motor Driver:** TB6612FNG Dual Motor Driver
- **Motors:** DC Geared Motors
- **Vehicle Model:** Two Miniature Vehicle Models
- **Power Supply:** Rechargeable Battery
- **Voltage Regulator:** DC-DC Buck Converter
- **Passive Components:** Resistors, Connecting Wires, and Breadboard
- **Programming Interface:** USB Type-C Cable
- **Chassis:** Vehicle Chassis for Mounting Components

2.4 Feasibility Study

2.4.1 Technical Feasibility

The proposed Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles is technically feasible because it uses readily available hardware components and widely supported software tools. The ESP32 microcontroller provides sufficient processing capabilities to interface with sensors, control the LED transmitter, process signals from the photodiode receiver, and display messages on an LCD. The high-brightness LED transmits data through modulated light signals, while the BPW34 photodiode and LM358 amplifier circuit detect and process the received optical signals. The MPU6050 accelerometer and flame sensor can be integrated to identify predefined emergency conditions and generate warning messages.

AES encryption can be implemented using a suitable software library to improve data confidentiality. The system can be programmed and tested using the Arduino IDE and Embedded C/C++. Since the components are commonly available and the prototype does not require complex network infrastructure, implementation and testing are practical for an academic project. However, successful communication depends on appropriate circuit design, signal processing, line-of-sight

availability, distance, and ambient lighting conditions. Therefore, the prototype is technically feasible for demonstrating short-range optical communication, although additional development and extensive testing would be necessary before considering deployment in real defence and military environments.

2.4.2 Robustness & Reliability

The proposed Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles is designed to provide reliable short-range communication through optical signals. The ESP32 microcontroller processes sensor data and manages message transmission and reception, while the high-brightness LED and BPW34 photodiode enable optical data exchange between vehicles. AES encryption can improve data confidentiality by protecting transmitted messages from unauthorized interpretation. The integration of the MPU6050 accelerometer and flame sensor supports emergency detection and alert generation based on predefined conditions. The LCD provides a convenient way to monitor received messages and emergency alerts.

The reliability of the system depends on the quality of the optical link, proper alignment between the transmitter and receiver, power supply stability, circuit connections, and environmental lighting conditions. Obstacles, excessive distance, and strong ambient light may reduce communication performance. Error detection, message validation, and retransmission mechanisms can further improve reliability. Although the prototype can demonstrate robust communication under controlled conditions, extensive testing under different environmental conditions is necessary before considering practical use in defence and military operations.

2.4.3 Economic Feasibility

The proposed Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles is economically feasible because it uses affordable and readily available electronic components. The system requires ESP32 microcontrollers, a high-brightness LED, a BPW34 photodiode, an LM358 amplifier module, an MPU6050 accelerometer, a flame sensor, an LCD, and supporting components. The required software tools, including Arduino IDE and suitable programming libraries, are available free of cost. The prototype can be assembled using basic electronic components and a simple vehicle model, reducing development and implementation expenses.

Furthermore, the modular design allows additional sensors and security features to be integrated gradually according to the available budget. Compared with developing a specialized military communication system from scratch, this project provides a cost-effective platform for academic research and experimental testing of optical communication. However, practical deployment in defence environments would require additional investment in rugged hardware, advanced encryption and authentication, environmental protection, and extensive reliability testing. Therefore, the proposed system is economically suitable for prototype development and educational purposes.

CHAPTER 3

ARCHITECTURAL DESIGN

3.1 Modules Design

The proposed Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles consists of the following modules, which work together to enable encrypted optical communication, emergency detection, and message display.

1. Sensor Monitoring Module: This module uses the MPU6050 accelerometer and a flame sensor to monitor vehicle movement and detect fire-related conditions. The ESP32 processes the sensor readings and identifies predefined emergency conditions to generate appropriate alert messages.

2. Data Processing Module: The ESP32 microcontroller processes sensor information, prepares the messages, and manages communication between the transmitter and receiver. It coordinates the operation of the sensors, encryption software, optical transmitter, and display unit.

3. Encryption and Decryption Module: This module uses the Advanced Encryption Standard (AES) algorithm to encrypt messages before transmission and decrypt them after reception. The transmitter and receiver must use compatible encryption settings and securely share the required cryptographic key.

4. Li-Fi Transmitter Module: This module uses a high-brightness LED to transmit digital information through modulated light signals. The ESP32 controls the LED according to the encoded data, enabling the optical transmission of emergency alerts and predefined messages.

5. Optical Receiver Module: This module uses a BPW34 photodiode and an LM358 amplifier circuit to detect incoming light signals and convert them into electrical signals. The receiving circuit conditions the signal so that the ESP32 can process the transmitted data.

6. Message Reception and Display Module: The receiving ESP32 processes the recovered data, verifies the message, and decrypts it when encryption is enabled. A 16×2 LCD with an I2C interface displays the received messages and emergency alerts for convenient monitoring.

3.2 Method & Algorithm Design

3.2.1 Assignment Method

The assignment method involves allocating specific tasks to the ESP32 microcontroller and associated hardware components to ensure proper system operation. The MPU6050 accelerometer and flame sensor provide vehicle movement and fire-related information to the ESP32, which

processes the sensor readings and generates appropriate emergency messages. The AES encryption module encrypts the messages before they are passed to the Li-Fi transmitter. The high-brightness LED converts the encrypted data into modulated optical signals, which are detected by the BPW34 photodiode at the receiving end. The LM358 amplifier conditions the received electrical signals, allowing the receiving ESP32 to recover and process the transmitted information.

The AES decryption module then decrypts the message using the appropriate cryptographic key, and the received information is displayed on the LCD. The power supply module provides the required operating voltages to the components. Each module is assigned a specific function, and all modules work together to demonstrate secure optical communication and sensor-based emergency alerts between vehicle models.

3.2.3 Overall Algorithm

· **Start:** Initialize the ESP32 microcontroller, sensors, LED transmitter, photodiode receiver, LCD, and required communication interfaces.
· **Initialize Sensors:** Configure the MPU6050 accelerometer and flame sensor to monitor vehicle movement and fire-related conditions.
· **Read Sensor Data:** Continuously collect sensor readings and evaluate them against predefined emergency conditions.
· **Generate Message:** Create an appropriate message containing sensor information, emergency alerts, or predefined communication data.
· **Encrypt Message:** Apply AES encryption to the message using a predefined cryptographic key to protect its confidentiality.
· **Encode Data:** Convert the encrypted message into a suitable sequence of digital bits for optical transmission.
· **Transmit Optical Signal:** Modulate the high-brightness LED according to the encoded data and transmit the information through light signals.
· **Receive Optical Signal:** Use the BPW34 photodiode to detect the incoming light signals and convert them into electrical signals.
· **Amplify and Process Signal:** Use the LM358 amplifier circuit to condition the received signal before passing it to the receiving ESP32 for processing.
· **Recover Transmitted Data:** The receiving ESP32 processes the electrical signal and reconstructs the transmitted digital message.
· **Decrypt Message:** Apply AES decryption using the corresponding cryptographic key to recover the original message.
· **Validate Message:** Check the received data for validity and integrity before accepting it for display or further processing.
· **Display Information:** Display the received message, sensor information, or emergency alert on the 16×2 LCD.

· **Monitor Continuously:** Continue monitoring sensor inputs and receiving messages to support ongoing communication between vehicles.
· **Stop:** End the process when the system is switched off or testing is completed.

3.3 Project Architecture

3.3.1 Architectural Diagram

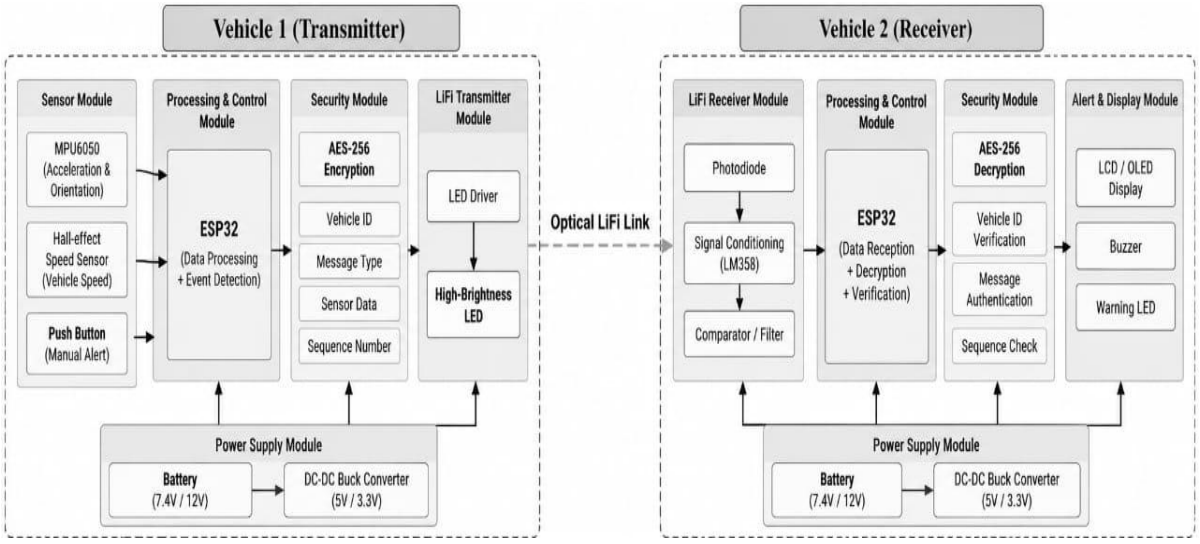


Figure 1: Architeture Diagram

The architecture of the proposed Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles consists of sensor monitoring, data processing, encryption, optical transmission, signal reception, decryption, and message display modules. The MPU6050 accelerometer and fire detection sensor collect information about vehicle movement and fire-related conditions and provide the data to the ESP32 microcontroller.

The ESP32 processes the sensor readings and generates appropriate emergency alerts or predefined messages. The AES encryption module encrypts the messages before they are transmitted through a high-brightness LED using modulated light signals. At the receiving vehicle, a BPW34 photodiode detects the optical signals, and an LM358 amplifier circuit conditions the received electrical signals for processing by the receiving ESP32. The receiver then reconstructs, validates, and decrypts the message using the corresponding cryptographic key.

Finally, the received information is displayed on a 16×2 LCD for monitoring. A regulated power supply provides the required power to the electronic components.

All modules work together to demonstrate encrypted optical communication and sensor-based emergency alerts between vehicle models. The system requires a suitable optical path and proper transmitter–receiver alignment for reliable communication.

3.3.2 Data Flow Diagram

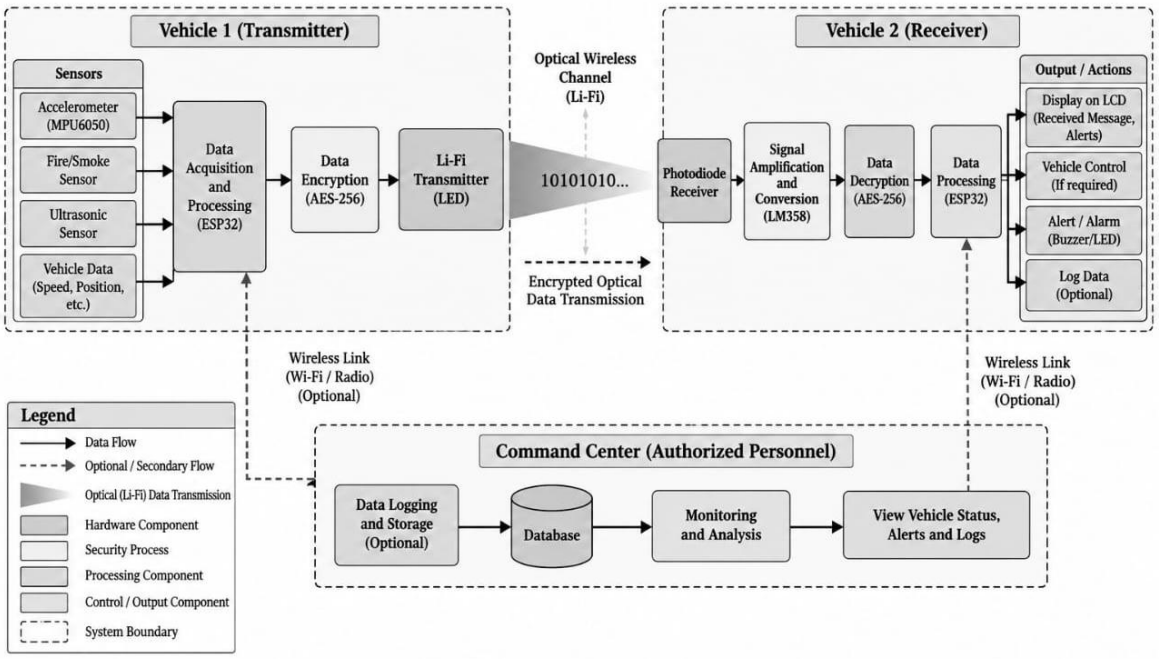


Figure 2: Data Flow Diagram

The Data Flow Diagram illustrates how information moves through the proposed secure optical V2V communication system. The MPU6050 accelerometer and fire sensor provide sensor readings to the transmitter ESP32, which processes the data and generates emergency messages. The AES encryption module encrypts the messages before the LED transmitter sends them through the Li-Fi optical channel. At the receiving vehicle, the BPW34 photodiode detects the light signals, and the LM358 amplifier conditions the received electrical signals. The receiving ESP32 recovers the transmitted data, and the AES decryption module restores the original message. Finally, the LCD displays the received emergency alerts and messages. This process demonstrates encrypted optical data transmission between two vehicle models.

3.3.3 Use Case Diagram

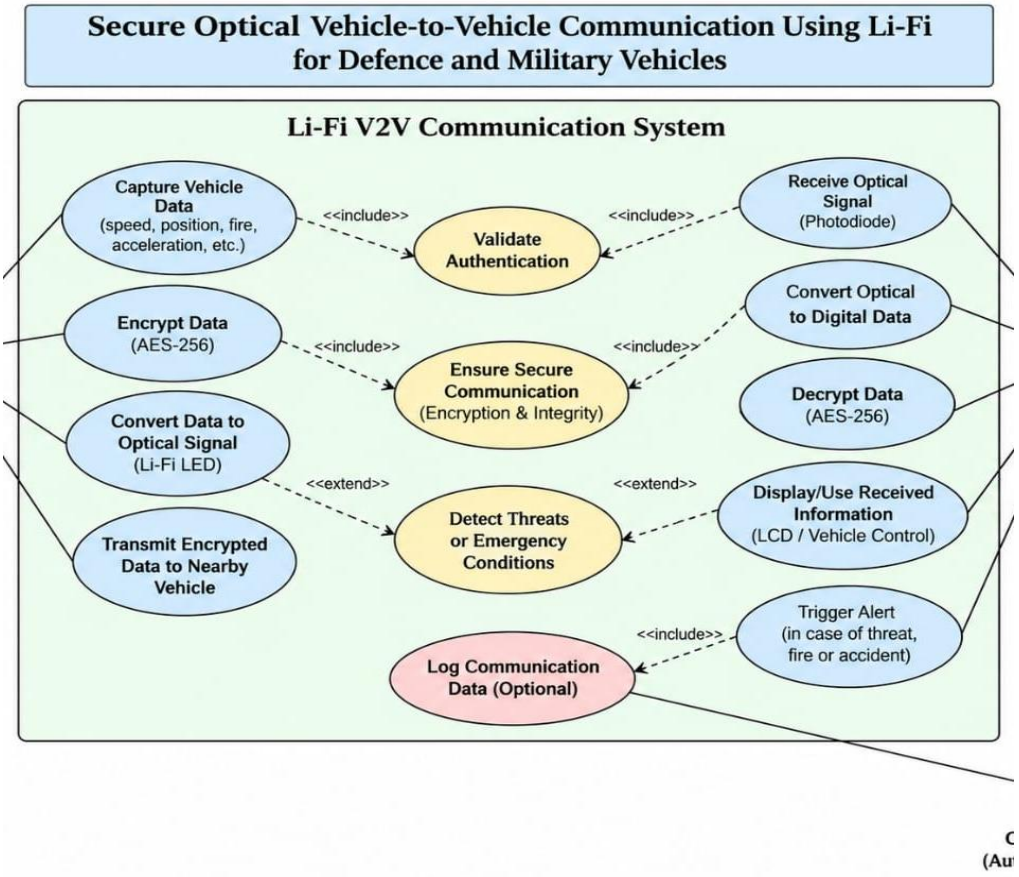


Figure 3: Uses Data diagram

The Data Flow Diagram is used to represent how data moves through the **Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles** system. It illustrates the flow of sensor information from the transmitter vehicle to the receiver vehicle, including data processing, AES encryption, optical transmission through an LED, signal detection using a photodiode, decryption, and displaying the received message on an LCD.

3.3.4 Sequence Diagram

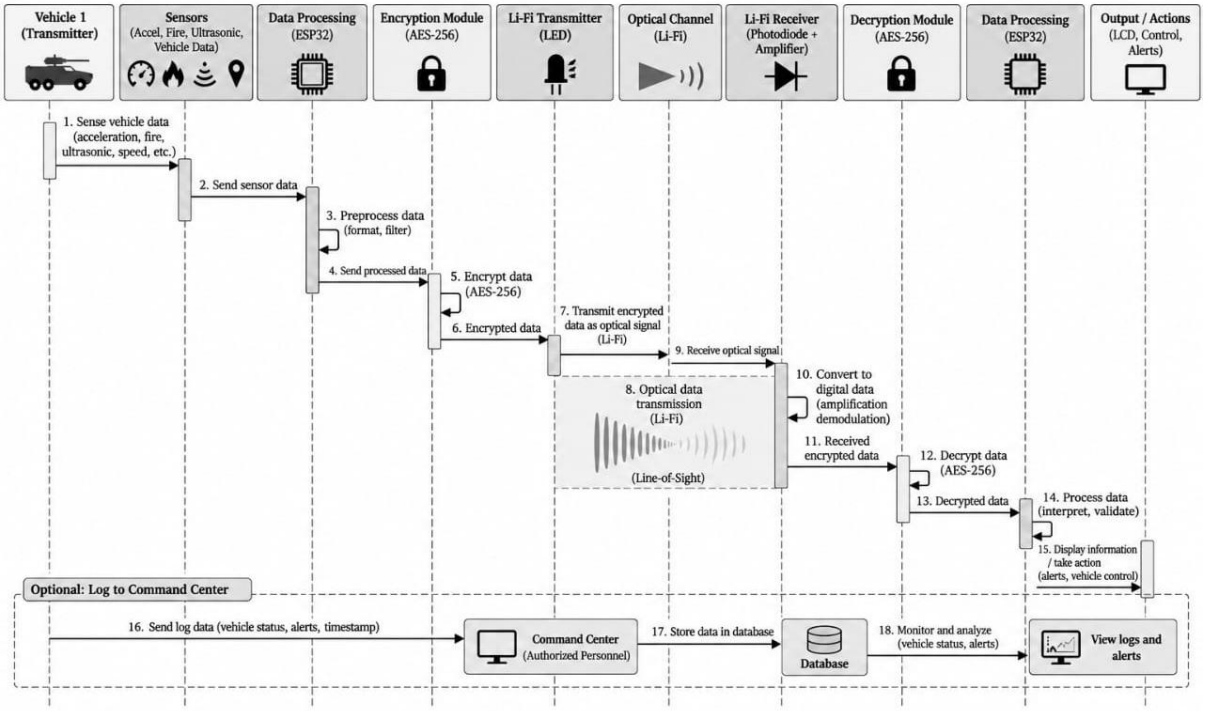


Figure 4: Sequence Diagram

The sequence diagram illustrates the interaction between the sensors, transmitter ESP32, AES encryption module, LED transmitter, photodiode receiver, receiving ESP32, and LCD display. Initially, the MPU6050 accelerometer and fire sensor provide sensor data to the transmitter ESP32. The ESP32 processes the information and generates an emergency message, which is encrypted using AES. The encrypted data is converted into optical signals by modulating a high-brightness LED. The photodiode at the receiving vehicle detects the transmitted light signals, and the LM358 amplifier conditions the resulting electrical signals. The receiving ESP32 recovers the transmitted data, decrypts the message using the corresponding key, and displays the received emergency alert or message on the LCD.

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3.3.5 Activity Diagram

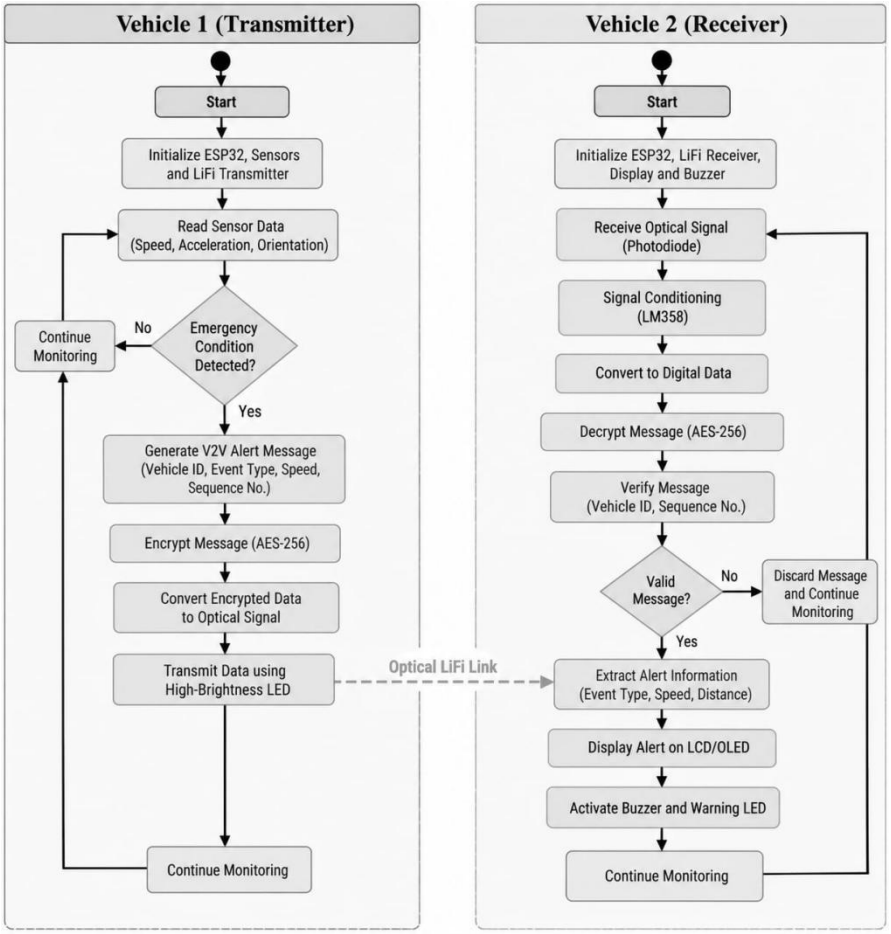


Figure 5: Activity Diagram

The process begins by initializing the ESP32 microcontroller, sensors, LED transmitter, optical receiver, and LCD. The system continuously reads data from the MPU6050 accelerometer and fire detection sensor to identify predefined emergency conditions. If an emergency is detected, the transmitter generates an alert; otherwise, it prepares a status update or predefined message. The message is encrypted using AES and transmitted through a modulated LED light signal. At the receiving vehicle, the photodiode detects the optical signal, and the receiver circuit conditions it for processing by the ESP32. The received data is recovered, decrypted, and validated before the message or emergency alert is displayed on the LCD. The system then returns to continuous monitoring until it is switched off.

CHAPTER 4

IMPLEMENTATION & TESTING

4.1 Coding Blocks

```
#include <Wire.h>
#include <WiFi.h>
#include <ThingSpeak.h>
#include <Preferences.h>
#include <LiquidCrystal_I2C.h>
#include "mbedtls/gcm.h"

#define RX_PIN 34

const char *WIFI_SSID = "wifi";
const char *WIFI_PASSWORD = "********";

unsigned long CHANNEL_ID = 1234567;
const char *WRITE_API_KEY = "rfgy7656hjgfgytr645yrth64684g";

WiFiClient client;

LiquidCrystal_I2C lcd(0x27, 16, 2);
Preferences prefs;

const unsigned char AES_KEY[32] = {
  0x10, 0x21, 0x32, 0x43,
  0x54, 0x65, 0x76, 0x87,
  0x98, 0xA9, 0xBA, 0xCB,
  0xDC, 0xED, 0xFE, 0x0F,
  0x11, 0x22, 0x33, 0x44,
  0x55, 0x66, 0x77, 0x88,
  0x99, 0xAA, 0xBB, 0xCC,
  0xDD, 0xEE, 0xF0, 0x01
};

const uint32_t SENDER_ID = 0x12345678;
```

```
const size_t PAYLOAD_SIZE = 7;
const size_t TAG_SIZE = 16;
const size_t SEQ_SIZE = 8;
const size_t FRAME_SIZE = SEQ_SIZE + PAYLOAD_SIZE + TAG_SIZE;

uint64_t lastSequence = 0;
unsigned long lastUploadTime = 0;

float acceleration = 0;
int16_t distanceCm = -1;
uint8_t messageFlags = 0;

void makeNonce(uint64_t seq, unsigned char nonce[12]) {
  nonce[0] = (SENDER_ID >> 24) & 0xFF;
  nonce[1] = (SENDER_ID >> 16) & 0xFF;
  nonce[2] = (SENDER_ID >> 8) & 0xFF;
  nonce[3] = SENDER_ID & 0xFF;

  for (int i = 0; i < 8; i++) {
    nonce[4 + i] = (seq >> (56 - 8 * i)) & 0xFF;
  }
}

uint64_t readUint64BE(const unsigned char *data) {
  uint64_t value = 0;

  for (int i = 0; i < 8; i++) {
    value = (value << 8) | data[i];
  }

  return value;
}

bool receiveFrame(unsigned char frame[FRAME_SIZE]) {
  // Search for the long start pulse.
  while (true) {
    unsigned long width = pulseIn(RX_PIN, HIGH, 50000);

    if (width == 0) {
      return false;
```

```
    }

    if (width >= 15000) {
      break;
    }
  }

  // Read all frame bits.
  for (size_t i = 0; i < FRAME_SIZE; i++) {
    frame[i] = 0;

    for (int bit = 7; bit >= 0; bit--) {
      unsigned long width = pulseIn(RX_PIN, HIGH, 15000);

      if (width == 0) {
        return false;
      }

      bool one = (width >= 5000);

      if (one) {
        frame[i] |= (1 << bit);
      }
    }
  }

  return true;
}

bool decryptFrame(
  const unsigned char frame[FRAME_SIZE],
  uint64_t &seq,
  float &outAcceleration,
  int16_t &outDistance,
  uint8_t &outFlags
) {
  unsigned char nonce[12];
  unsigned char plaintext[PAYLOAD_SIZE];

  seq = readUint64BE(frame);
```

```
if (seq == 0 || seq <= lastSequence) {
  Serial.println("Rejected old or replayed frame.");
  return false;
}

const unsigned char *ciphertext = frame + SEQ_SIZE;
const unsigned char *tag =
  frame + SEQ_SIZE + PAYLOAD_SIZE;

makeNonce(seq, nonce);

mbedtls_gcm_context ctx;
mbedtls_gcm_init(&ctx);

int result = mbedtls_gcm_setkey(
  &ctx,
  MBEDTLS_CIPHER_ID_AES,
  AES_KEY,
  256
);

if (result == 0) {
  result = mbedtls_gcm_auth_decrypt(
    &ctx,
    PAYLOAD_SIZE,
    nonce,
    sizeof(nonce),
    nullptr,
    0,
    tag,
    TAG_SIZE,
    ciphertext,
    plaintext
  );
}

mbedtls_gcm_free(&ctx);

if (result != 0) {
```

```
    Serial.println("AUTHENTICATION FAILED: message rejected.");
    return false;
  }

  memcpy(&outAcceleration, plaintext, sizeof(float));
  memcpy(&outDistance, plaintext + 4, sizeof(int16_t));
  outFlags = plaintext[6];

  // Persist only after authentication succeeds.
  lastSequence = seq;
  prefs.putULong64("rxseq", lastSequence);

  return true;
}

int getMessageCode() {
  bool fire = messageFlags & (1 << 0);
  bool impact = messageFlags & (1 << 1);
  bool obstacle = messageFlags & (1 << 2);
  bool sensorError = messageFlags & (1 << 3);

  if (sensorError) return 5;

  int alerts = fire + impact + obstacle;

  if (alerts > 1) return 4;
  if (fire) return 1;
  if (obstacle) return 2;
  if (impact) return 3;

  return 0;
}

const char *getMessageText() {
  switch (getMessageCode()) {
    case 1: return "FIRE DETECTED";
    case 2: return "OBSTACLE";
    case 3: return "POSSIBLE IMPACT";
    case 4: return "MULTIPLE ALERTS";
    case 5: return "SENSOR ERROR";
```

```
    default: return "NORMAL";
  }
}

void updateLCD() {
  lcd.setCursor(0, 0);
  lcd.print("                ");
  lcd.setCursor(0, 0);
  lcd.print(getMessageText());

  lcd.setCursor(0, 1);
  lcd.print("                ");
  lcd.setCursor(0, 1);

  if (distanceCm >= 0) {
    lcd.print("Dist:");
    lcd.print(distanceCm);
    lcd.print("cm");
  } else {
    lcd.print("Dist: No Echo");
  }
}

void uploadThingSpeak() {
  if (WiFi.status() != WL_CONNECTED) {
    Serial.println("Wi-Fi unavailable; skipping upload.");
    return;
  }

  if (millis() - lastUploadTime < 20000) {
    return;
  }

  ThingSpeak.setField(1, acceleration);
  ThingSpeak.setField(2, (messageFlags & 1) ? 1 : 0);
  ThingSpeak.setField(3, distanceCm);
  ThingSpeak.setField(4, getMessageCode());

  int result = ThingSpeak.writeFields(CHANNEL_ID, WRITE_API_KEY);
```

```
  if (result == 200) {
    Serial.println("Verified data uploaded to ThingSpeak.");
  } else {
    Serial.printf("ThingSpeak error: %d\n", result);
  }

  lastUploadTime = millis();
}

void setup() {
  Serial.begin(115200);

  pinMode(RX_PIN, INPUT);

  Wire.begin(21, 22);

  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Secure V2V");
  lcd.setCursor(0, 1);
  lcd.print("Starting...");

  prefs.begin("securev2v", false);
  lastSequence = prefs.getULong64("rxseq", 0);

  WiFi.mode(WIFI_STA);
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

  unsigned long start = millis();

  while (WiFi.status() != WL_CONNECTED &&
         millis() - start < 15000) {
    delay(500);
  }

  ThingSpeak.begin(client);

  if (WiFi.status() == WL_CONNECTED) {
    Serial.println("Wi-Fi connected.");
```

```
  } else {
    Serial.println("Wi-Fi unavailable; local reception continues.");
  }

  lcd.clear();
  lcd.print("Waiting for");
  lcd.setCursor(0, 1);
  lcd.print("Li-Fi message");
}

void loop() {
  unsigned char frame[FRAME_SIZE];

  if (!receiveFrame(frame)) {
    return;
  }

  uint64_t seq;
  float receivedAcceleration;
  int16_t receivedDistance;
  uint8_t receivedFlags;

  if (!decryptFrame(
        frame,
        seq,
        receivedAcceleration,
        receivedDistance,
        receivedFlags
      )) {
    lcd.setCursor(0, 0);
    lcd.print("                ");
    lcd.setCursor(0, 0);
    lcd.print("INVALID MESSAGE");
    Serial.println("Message not displayed as trusted data.");
    return;
  }

  // Update the displayed and uploaded values only after authentication.
  acceleration = receivedAcceleration;
  distanceCm = receivedDistance;
```

```
  messageFlags = receivedFlags;

  Serial.printf(
    "Verified sequence=%llu, acceleration=%.2f, distance=%d, flags=%u\n",
    (unsigned long long)seq,
    acceleration,
    distanceCm,
    messageFlags
  );

  updateLCD();
  uploadThingSpeak();
}
```

4.2 Test Case Study

4.2.1 Test Case 1

Test Scenario: Verify sensor initialization.

Objective: Ensure that the ESP32 successfully initializes the MPU6050 accelerometer and fire detection sensor.

Preconditions:

- ESP32 is connected to the sensors.
- The power supply is connected properly.
- The required software is uploaded to the ESP32.

Test Steps:

1. Switch on the system.
2. Initialize the ESP32 and connected sensors.
3. Read the sensor values through the Serial Monitor.

Expected Result:

- The sensors should initialize successfully.
- Sensor readings should be available for processing.

Actual Result: To be recorded during hardware testing.

Status: Pending testing.

4.2.2 Test Case 2

Test Scenario: Verify emergency detection.

Objective: Ensure that the system generates an appropriate alert when a predefined emergency condition is detected.

Preconditions:

- The ESP32 and sensors are operational.

- Emergency detection thresholds are configured.
- The transmitter program is uploaded.

Test Steps:

1. Switch on the transmitter system.
2. Trigger a predefined test condition for the accelerometer or fire sensor.
3. Observe the sensor readings and generated message.

Expected Result:

- The sensor should detect the configured condition.
- The ESP32 should generate the corresponding emergency message.

Actual Result: To be recorded during hardware testing.

Status: Pending testing.

4.2.3 Test Case 3

Test Scenario: Verify AES encryption.

Objective: Ensure that emergency messages are encrypted before being transmitted through the Li-Fi channel.

Preconditions:

- The AES encryption library is configured.
- The encryption key is available.
- The transmitter program is operational.

Test Steps:

1. Generate a predefined message.
2. Pass the message to the AES encryption module.
3. Observe the encrypted output through the Serial Monitor.

Expected Result:

- The original message should be encrypted successfully.
- The encrypted output should be ready for transmission.

Actual Result: To be recorded during hardware testing.

Status: Pending testing.

4.2.4 Test Case 4

Test Scenario: Verify optical data transmission.

Objective: Ensure that the LED transmits the encrypted message using modulated light signals.

Preconditions:

- The transmitter ESP32 and LED circuit are connected properly.
- The encryption module is operational.
- A suitable optical path is available.

Test Steps:

1. Switch on the transmitter.
2. Generate and encrypt a test message.

3. Transmit the encoded data through the LED.
4. Observe the transmitter operation.

Expected Result:

- The LED should transmit the encoded data according to the implemented modulation protocol.
- The optical signal should be detectable by the receiver under suitable conditions.

Actual Result: To be recorded during hardware testing.

Status: Pending testing.

4.2.5 Test Case 5

Test Scenario: Verify optical signal reception.

Objective: Ensure that the photodiode and LM358 amplifier circuit receive and condition the transmitted optical signal.

Preconditions:

- The BPW34 photodiode and LM358 circuit are connected properly.
- The receiver ESP32 is powered on.
- The transmitter and receiver are suitably aligned.

Test Steps:

1. Switch on both transmitter and receiver.
2. Transmit a predefined encrypted message.
3. Observe the receiver circuit output.
4. Check whether the receiving ESP32 recovers the transmitted data.

Expected Result:

- The photodiode should detect the modulated light signal.
- The receiver circuit should produce a suitable signal for digital data recovery.

Actual Result: To be recorded during hardware testing.

Status: Pending testing.

CHAPTER 5

RESULTS

5.1 Resulting Screens

5.1.1 Working Model

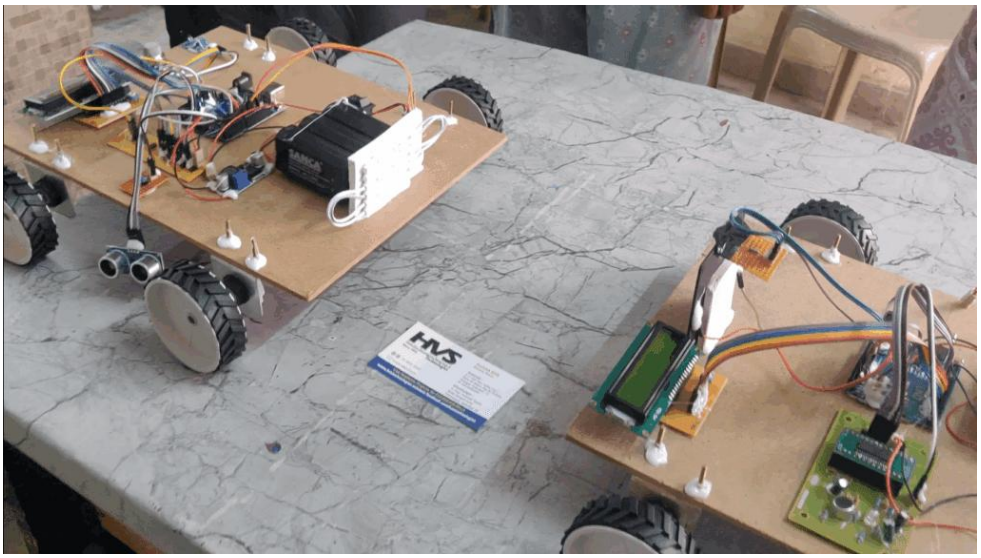


Figure 6: Working Model

5.1.2 Dashboard

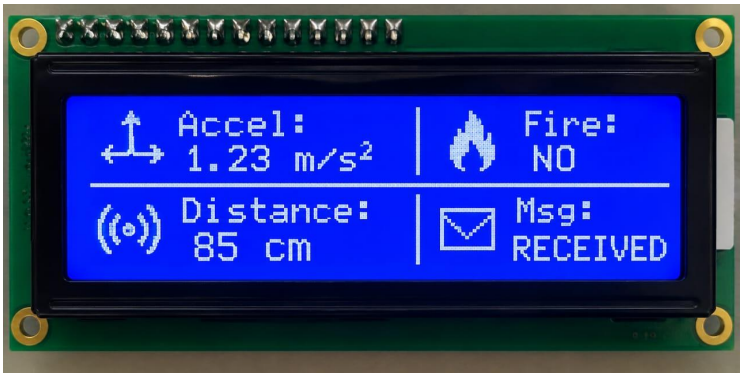


Figure 7: Dashboard

5.1.3 Dashboard

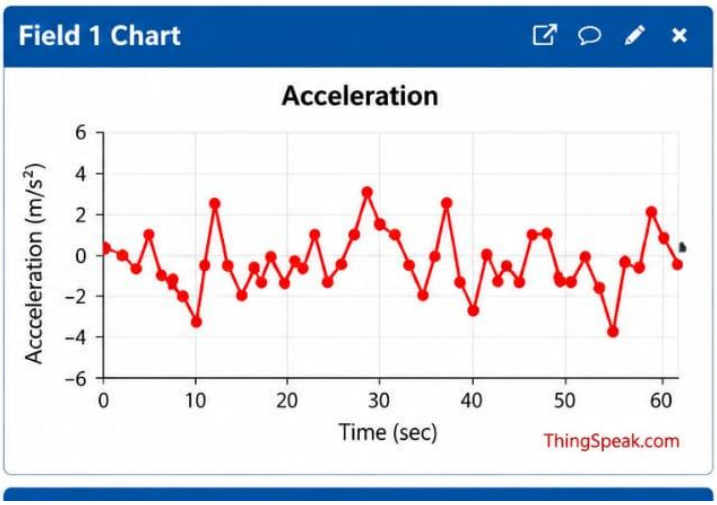
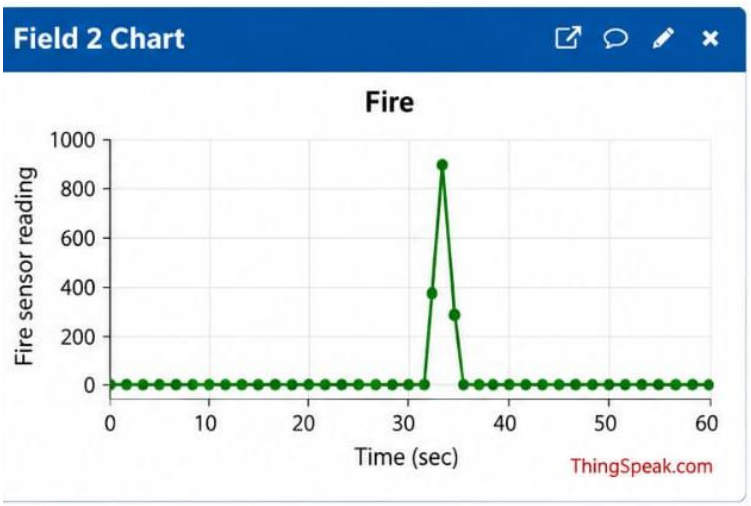


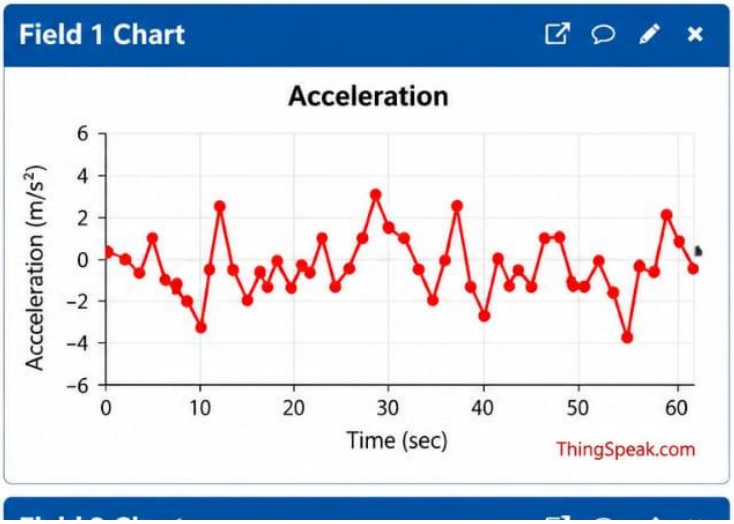
Figure 8: Dashboard

5.3 Resulting graphs

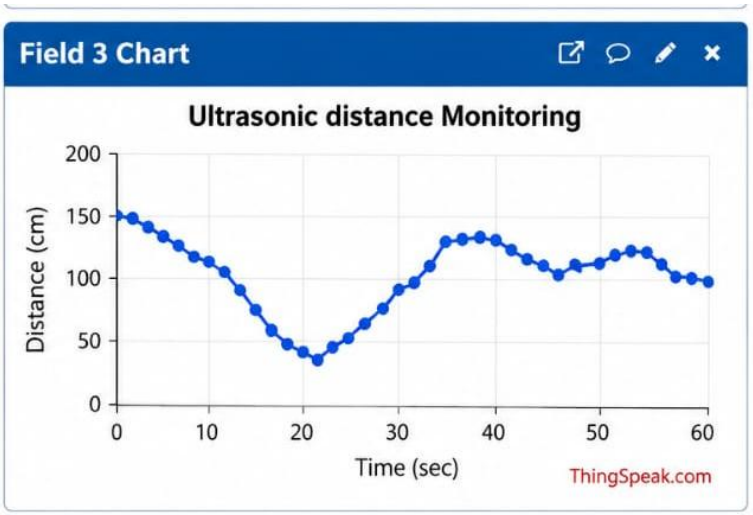
5.3.1 Result graph 1:



5.3.2 Result graph 2:



5.3.3 Result graph 3:



5.4 Results Analysis

The results analysis of the proposed Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles focuses on evaluating the system's functionality, communication performance, and data security. The system is designed to collect sensor readings using the MPU6050 accelerometer and fire detection sensor, process the information through the ESP32 microcontroller, and generate appropriate emergency alerts. AES encryption is used to protect messages before they are transmitted through the Li-Fi channel using a high-brightness LED. At the receiving end, the BPW34 photodiode detects the optical signals, and the LM358 amplifier circuit conditions the received signals for processing by the receiving ESP32.

The encrypted message is then decrypted and displayed on the LCD. The system's performance can be evaluated by checking successful message transmission, correct decryption, sensor response, and emergency alert display. Additional tests can assess the effects of communication distance, optical alignment, and ambient lighting on message reception. The expected outcome is a prototype capable of demonstrating encrypted optical message transmission and sensor-based emergency notification under suitable operating conditions. Actual performance, including transmission accuracy, communication range, and response time, must be established through experimental testing. The analysis can help identify system limitations and suggest improvements in signal processing, error detection, encryption, and overall reliability.

5.4.1 Results Summary

The proposed **Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles** aims to demonstrate secure, short-range communication between two vehicles using optical signals. The system integrates ESP32 microcontrollers, a high-brightness LED transmitter, a BPW34 photodiode receiver, an LM358 amplifier module, an MPU6050 sensor, a fire detection sensor, and an LCD display. Sensor information and emergency messages are processed by the transmitting ESP32 and can be encrypted using the AES algorithm before being transmitted through the Li-Fi channel. At the receiving end, the optical signals are converted into electrical signals, processed by the receiver ESP32, decrypted, and displayed on the LCD. The proposed system is expected to demonstrate sensor-based emergency alerts and encrypted message transmission under suitable operating conditions. Its performance depends on optical alignment, communication distance, ambient lighting, and successful encryption and decryption. Actual results, including transmission accuracy, response time, and maximum communication range, must be confirmed through practical testing. Overall, the project provides a prototype for exploring optical wireless communication as a complementary technology for defence vehicle coordination, with opportunities for further improvement in security, reliability, and communication performance.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The project “Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles” proposes an optical wireless communication system for exchanging information and emergency alerts between vehicles. The system integrates ESP32 microcontrollers, a high-brightness LED transmitter, a BPW34 photodiode receiver, an LM358 amplifier module, an MPU6050 accelerometer, a fire detection sensor, and an LCD display. By transmitting data through modulated light signals, the system provides an alternative to conventional radio-frequency communication for short-range, line-of-sight applications.

AES encryption can be incorporated to protect messages from unauthorized disclosure during transmission. Sensor information can be used to generate alerts, which are received, processed, and displayed on the receiving vehicle's LCD. The proposed approach offers potential advantages, including reduced dependence on radio-frequency links, localized optical communication, and real-time emergency message exchange. However, its effectiveness depends on factors such as communication distance, LED and photodiode alignment, ambient lighting, and obstacles in the optical path.

6.2 Future Scope

The proposed Secure Optical V2V Communication Using Li-Fi for Defence and Military Vehicles can be improved in several ways to enhance its security, reliability, and practical applications. In the future, the system can be extended to support **bidirectional communication**, allowing both vehicles to transmit and receive information. Advanced encryption and message authentication techniques can be implemented to improve data confidentiality and integrity. The communication range and reliability can be enhanced by using high-speed LEDs, improved photodiode receiver circuits, and advanced signal-processing techniques. Additional sensors, such as GPS modules, temperature sensors, and collision detection sensors, can be integrated to provide vehicle location, environmental information, and early warnings about potential hazards. A real-time monitoring dashboard can also be developed to display vehicle status, emergency alerts, and communication information. Furthermore, error detection, retransmission mechanisms, and adaptive signal control can be introduced to improve communication under varying lighting conditions. The system can be tested in different environmental conditions and scaled to support communication among multiple vehicles. For practical defence applications, the prototype would require extensive field testing, security evaluation, and integration with existing communication systems. These improvements could help develop a more reliable and secure optical communication solution for military convoys, emergency response vehicles, and other specialized transportation applications..

REFERENCES

1. IEEE, IEEE Standard for *Local and Metropolitan Area Networks—Part 15.7: Short-Range Optical Wireless Communications,* IEEE Std. 802.15.7-2018, 2018.

2. National Institute of Standards and Technology (NIST), Advanced Encryption Standard (AES), Federal Information Processing Standards Publication 197 (FIPS 197), updated May 2023.

3. IEEE, "IEEE 802.11p Standard—*Wireless Access in Vehicular Environments (WAVE)," IEEE Technology Navigator.*

4. Vishay Semiconductors, BPW34, BPW34S: Silicon PIN Photodiode—Data Sheet.

5. Espressif Systems, ESP32 Series Datasheet, Espressif Systems.

6. TDK InvenSense, *MPU-6000 and MPU-6050 Product Specification*, Revision 3.4.

7. Texas Instruments, LM358 Dual Operational Amplifier Data Sheet, Texas Instruments.

8. Arduino, Arduino IDE Documentation, Arduino Documentation.

9. Adafruit Industries, *Adafruit MPU6050 Library and Guide, Adafruit Learning System.*

10. IEEE, "Vehicle-to-Vehicle Communication," IEEE Technology Navigator.

POSTER PRESENTATION

